No.

問1

$z = re^{i\theta}$ とおくと $d\mu = \frac{1}{\pi} e^{-r^2} \cdot r\, d\theta\, dr$ となるから

$$(z^n, z^m) = \frac{1}{\pi}\int_0^{2\pi} e^{i(m-n)\theta}\, d\theta \cdot \int_0^{\infty} r^{m+n+1} e^{-r^2} dr$$

$$= 2\delta_{m,n} \cdot \int_0^{\infty} r^{2n+1} e^{-r^2} dr = \delta_{m,n} \cdot n! \quad \cdots (4)$$

これより

$$(f, g) = \left(\sum_n f_n z^n, \sum_m g_m z^m\right) = \sum_n \sum_m f_n^* g_m (z^n, z^m)$$

$$= \sum_n f_n^* g_n \cdot n! \quad \cdots (6)$$

特に $\|f\|^2 = (f, f) = \sum_n n! |f_n|^2 \quad \cdots (5)$

No.

問題 2

$$\left[\{f_n(z) = \sum_{k=0}^{\infty} a_{n,k} z^k\}_{n=1}^{\infty} \in \mathcal{F} \right]_{(A)} \text{ and } \left[\|f_n - f_m\| \to 0 \quad (n, m \to \infty) \right]_{(B)}$$

$$\Longleftrightarrow \left[\forall_{n=1,2,\cdots}, \sum k! |a_{n,k}|^2 < C_n < \infty \right]_{(A')} \text{ and } \left[\lim_{n,m\to\infty} \sum k! |a_{n,k} - a_{m,k}|^2 = 0 \right]_{(B')}$$

$$(A') \Rightarrow \sum k! \lim_{n\to\infty} (a_{n,k}^* a_{n,k}) < C_\infty < \infty \quad \cdots (a)$$

$$(B') \Rightarrow \sum k! \lim_{n,m\to\infty} (a_{n,k} - a_{m,k})^* (a_{n,k} - a_{m,k})$$

$$\Rightarrow \sum k! \cdot \left\{ \lim_{n\to\infty} |a_{n,k}|^2 + \lim_{m\to\infty} |a_{m,k}|^2 - \lim_{n,m\to\infty} (a_{n,k}^* a_{m,k} + a_{m,k}^* a_{n,k}) \right\} = 0 \quad \cdots (b)$$

(a) and (b) $\Rightarrow \sum k! \cdot \lim_{n,m\to\infty} (a_{n,k}^* a_{m,k} + a_{m,k}^* a_{n,k}) < 2C_\infty$

$$\Rightarrow \sum k! \left\{ (\lim_{n\to\infty} a_{n,k})^* (\lim_{m\to\infty} a_{m,k}) + (\lim_{m\to\infty} a_{m,k})^* (\lim_{n\to\infty} a_{n,k}) \right\} < 2C_\infty$$

$$\Longleftrightarrow \sum k! \left| \lim_{n\to\infty} a_{n,k} \right|^2 < C_\infty$$

$\therefore$ (A) and (B) $\Rightarrow \|f\| < \sqrt{C_\infty} < \infty$ 」

やっぱりこの手の… よくわかりません！

COOL

No.

問3

$\forall c, z \quad f(cz) = c^\lambda f(z)$ ... (10)

$\Rightarrow \forall c, z \quad \frac{\partial}{\partial c} f(cz) = \lambda c^{\lambda-1} f(z)$

$\Rightarrow \forall c \neq 0, z \quad c \frac{\partial}{\partial c} f(cz) = \lambda f(cz)$

$\Rightarrow \forall c \quad c \frac{\partial}{\partial c} f(c) = \lambda f(c)$ .... (9)

$f(z) = \sum_{n=0}^{\infty} f_n z^n$ と表すと

$\forall z \quad z \frac{\partial}{\partial z} \sum_{n=0}^{\infty} f_n z^n = \lambda \sum_{n=0}^{\infty} f_n z^n$ ... (9)

$\Rightarrow \forall z, \quad \sum_{n=0}^{\infty} (n-\lambda) f_n z^n = 0$

$\Rightarrow \forall n, \quad (n-\lambda) f_n = 0$

$\Leftarrow \forall n, c \quad (c^n - c^\lambda) f_n = 0$ ☺ c^n は一価関数

$\Leftrightarrow \sum_{n=0}^{\infty} f_n (cz)^n = c^\lambda \sum_{n=0}^{\infty} f_n z^n$

∴ (10) ⇒ (9) and (9) ⇒ (10) すなわち (9) ⇔ (10)

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No.

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問4

$$zf = \sum_{n=0}^{\infty} f_n z^{n+1} = \sum_{n=1}^{\infty} f_{n-1} z^n$$

$$\frac{\partial}{\partial z} g = \sum_{n=1}^{\infty} n g_n z^{n-1} = \sum_{n=0}^{\infty} (n+1) g_{n+1} z^n$$

$$(zf, g) - \left(f, \frac{\partial}{\partial z} g\right) = \sum_{n=1}^{\infty} n!\, f_{n-1}^{*} g_n - \sum_{n=0}^{\infty} n!\, f_n^{*} (n+1) g_{n+1}$$

$$= \sum_{n=1}^{\infty} n!\, f_{n-1}^{*} g_n - \sum_{n=0}^{\infty} (n+1)!\, f_n^{*} g_{n+1}$$

$$= 0 \quad //$$

No.

P5| 5

$$e_a(z) = e^{a^* z} = \sum_{n=0}^{\infty} \frac{1}{n!} (a^*)^n z^n$$

$$(e_a, f) = \sum_{n=0}^{\infty} a^n f_n = f(a)$$

No.

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Pg 6

$(e_a, e_b) = e_b(a) = e^{b^* a}$, $\|e_a\|^2 = (e_a, e_a) = e^{a^* a}$

$\forall f(z) = \sum_{n=0}^{\infty} f_n z^n$

$|f(z)|^2 = |f_0 + f_1 z + f_2 z^2 + \cdots|^2$

$\leq (|f_0| + |f_1||z| + |f_2||z|^2 + \cdots)^2$ ☹ 三角不等式

$\leq (|f_0|^2 + |f_1|^2 + 2!|f_2|^2 + \cdots)(1 + |z|^2 + \frac{|z|^4}{2!} + \cdots)$ ☹ シュワルツの不等式

$= \|f\|^2 \cdot \|e_z\|^2 \leq \|f\|^2 (e^{\frac{1}{2} z^* z})^2$

$\|f\|^2 = \int d\mu |f(z)|^2 \leq \int d\mu\, c^2 e^{\alpha |z|^2}$ ☹ $|f(z)| \leq c e^{\frac{1}{2}\alpha z^* z}$

$= \frac{c^2}{\pi} \int r e^{-r^2} e^{\alpha r^2} dr d\theta$ ☹ $d\mu = \frac{1}{\pi} e^{-r^2} r dr d\theta$

$= \frac{c^2}{\pi} \cdot \frac{1}{2(1-\alpha)} \cdot 2\pi = \frac{c^2}{1-\alpha} < \infty$ J

No.

問題7

$[a^+, a^+] = 0$, $[a, a] = 0$ は自明

$$[a, a^+] = aa^+ - a^+a = \frac{1}{2}\left(q + \frac{\partial}{\partial q}\right)\left(q - \frac{\partial}{\partial q}\right) - \frac{1}{2}\left(q - \frac{\partial}{\partial q}\right)\left(q + \frac{\partial}{\partial q}\right)$$

$$= -q\frac{\partial}{\partial q} + \frac{\partial}{\partial q}q = 1$$

$$\left[-\frac{1}{2}\left(\frac{\partial}{\partial q}\right)^2 + \frac{q^2}{2}\right] = a^+a + \frac{1}{2} \quad \text{(1)}$$

$$a^+a\,|n\rangle = n\,|n\rangle$$

ここで $(a^+a)\,a^+|n\rangle = \left([a^+a, a^+] + a^{+2}a\right)|n\rangle = (n+1)\,a^+|n\rangle$ (2)

$$a^+|n\rangle = c\,|n+1\rangle \qquad \langle n|aa^+|n\rangle = |c|^2 \rightarrow c = 1/\sqrt{n+1}$$

よって $|n\rangle = \frac{1}{\sqrt{n}}a^+|n-1\rangle = \cdots = \frac{1}{\sqrt{n!}}(a^+)^n|0\rangle$ //

No.

問 8

$$\int d\mu\, |z\rangle f(z) = \int d\mu\, e^{z^* a^\dagger} |0\rangle f(z) = \sum_{n=0}^{\infty} \int d\mu\, e^{z^* a^\dagger} |0\rangle f_n z^n$$

$$= \sum_n \sum_m \int d\mu \frac{1}{m!} (z^*)^m (a^\dagger)^m |0\rangle f_n z^n$$

$$= \sum_n \sum_m \int d\mu \frac{1}{m!} (z^*)^m |m\rangle f_n z^n$$

$$= \sum_n \int d\mu \frac{1}{n!} (z^*)^n |n\rangle f_n z^n$$ ☺ 問 1 (4) より

$$= \sum_n \int d\mu\, f_n (a^\dagger)^n |0\rangle = f(a^\dagger)|0\rangle \quad //$$

$$\int d\mu\, |z\rangle\langle z| = \int d\mu \left(\sum_n |n\rangle \frac{(z^*)^n}{\sqrt{n!}} \right) \cdot \left(\sum_m \frac{(z)^m}{\sqrt{m!}} \langle m| \right)$$

$$= \int d\mu \sum_n |n\rangle\langle n| \frac{(z^*)^n (z)^n}{n!}$$ ☺ (4) より

$$= \sum_n |n\rangle\langle n| = 1 \quad //$$

No.

Prob 9

$$e^{z^*(a^\dagger - a)} = e^{-\frac{1}{2}z^{*2}} e^{z^* a^\dagger} e^{-z^* a} \quad ☺ \ [z^* a^\dagger, -z^* a] = z^{*2}$$

$$\langle \beta | z \rangle = \langle \beta | e^{z^* a^\dagger} | 0 \rangle = \langle \beta | e^{z^* a^\dagger} e^{-z^* a} | 0 \rangle \quad ☺ \ e^{-z^* a}|0\rangle = |0\rangle$$

$$= e^{\frac{1}{2}z^{*2}} \langle \beta | e^{z^*(a^\dagger - a)} | 0 \rangle$$

$$= e^{\frac{1}{2}z^{*2}} e^{-\sqrt{2}z^* \frac{\partial}{\partial \beta}} \langle \beta | 0 \rangle$$

$$= e^{\frac{1}{2}z^{*2}} \left(1 - \sqrt{2}z^* \frac{\partial}{\partial \beta} + z^{*2} \frac{\partial^2}{\partial \beta^2} + \cdots \right) X_0(\beta)$$

$$= e^{\frac{1}{2}z^{*2}} X_0(\beta - \sqrt{2}z^*)$$

$$= \pi^{-\frac{1}{4}} e^{\frac{1}{2}z^{*2}} e^{-\frac{1}{2}(\beta - \sqrt{2}z^*)^2}$$

$$= \pi^{-\frac{1}{4}} e^{-\frac{1}{2}(\beta^2 - 2\sqrt{2}\beta z^* + z^*)}$$

$$= \pi^{-\frac{1}{4}} e^{-\frac{1}{2}(\beta - \sqrt{2}z^*)^2} = A(\beta, z)^*$$

p.10

$$\int_{-\infty}^{\infty} dq\, A(q,z)\,A(q,z')^* = \frac{1}{\sqrt{\pi}}\int_{-\infty}^{\infty} dp\; e^{-\left(q-\frac{z+z'^*}{\sqrt{2}}\right)^2 + zz'^*}$$

$$= e^{zz'^*} \qquad \cdots (39)$$

$$\int_{-\infty}^{\infty} dq\, A(q,z)\,\Phi(q) = \int_{-\infty}^{\infty} dq\, A(q,z) \int d\mu(z')\, A(q,z')^* f(z')$$

$$= \int d\mu(z')\, f(z')\, e^{zz'^*} \qquad \because (39)$$

$$= f(z) \qquad \because (17)$$

P.9 11

$$\langle \xi | z \rangle = \langle \xi | \sum_{n=0}^{\infty} |n\rangle \frac{(z^*)^n}{\sqrt{n!}}$$ ☺ (31)式

$$= \sum_{n=0}^{\infty} \langle \xi | n \rangle \cdot \left(\frac{z^n}{\sqrt{n!}} \right)^* = \sum_{n=0}^{\infty} X_n(\xi) \cdot U_n(z)^*$$ ☺ (8)式

$$A(\xi, z)^* = \pi^{-\frac{1}{4}} e^{-\frac{1}{2}\xi^2} e^{-\frac{1}{2}z^{*2} + \sqrt{2}\,\xi z^*}$$

$$= \pi^{-\frac{1}{4}} e^{-\frac{1}{2}\xi^2} \sum_{n=0}^{\infty} H_n(\xi) \frac{(z^*/\sqrt{2})^n}{n!}$$ ☺ (42)式で $t \to \frac{z^*}{\sqrt{2}}$, $x \to \xi$

(41)式より $$= \sum_{n=0}^{\infty} X_n(\xi) \frac{(z^*)^n}{\sqrt{n!}}$$

$$\therefore \quad X_n(\xi) = \pi^{-\frac{1}{4}} e^{-\frac{1}{2}\xi^2} \cdot 2^{-\frac{n}{2}} H_n(\xi) \quad //$$

$$\int d\mu\, A(\xi, z)\, A(\xi', z)^* = \int dx\,dy \frac{1}{\pi} \cdot e^{-z^* z} \cdot \frac{1}{\sqrt{\pi}} e^{-\frac{1}{2}\xi^2 + \sqrt{2}\,\xi z - \frac{1}{2}z^2} e^{-\frac{1}{2}\xi'^2 + \sqrt{2}\,\xi' z^* - \frac{1}{2}z^{*2}}$$

$$= \pi^{-3/2} \int dx\,dy\; e^{-\frac{\xi^2}{2} - \frac{\xi'^2}{2}} e^{\sqrt{2}(\xi z + \xi' z^*)} e^{-\frac{1}{2}(z + z^*)^2}$$

$$= \pi^{-3/2} e^{-\frac{\xi^2}{2} - \frac{\xi'^2}{2}} \int dy\, e^{\sqrt{2}(\xi - \xi')iy} \int dx\, e^{-2x^2} \qquad (z = x + iy)$$

$$= \pi^{-3/2} e^{-\frac{\xi^2}{2} - \frac{\xi'^2}{2}} \cdot \frac{2\pi}{\sqrt{2}} \delta(\xi - \xi') \cdot \sqrt{\frac{\pi}{2}} \qquad y' = \sqrt{2}\,y$$

$$= e^{-\frac{1}{2}(\xi^2 + \xi'^2)} \delta(\xi - \xi')$$

No.

P.N 12

$$\Psi(\xi) = \int d\mu\, A(\xi, z)^* f(z)$$

$$= \sum_{n=0}^{\infty} X_n(\xi) \int d\mu\, U_n(z)^* f(z)$$